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Anisotropy Sensitivity in Ultra-Deep Azimuthal Resistivity Technologies

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Logging While Drilling (LWD) tools can be used to measure formation resistivity from Depths of Investigation (DOI) of inches out to dozens of feet. Using multi-depth, multi-frequency electromagnetic propagation resistivity (EM) tools, resistivity profiles can be inverted to provide distance to resistivity boundaries and petrophysical information. The primary piece of information produced is the horizontal resistivity map (R_h), but that is only one piece of the puzzle. The other pieces are the vertical resistivity (R_v), layer thickness defined by boundary position and the dip of the formation. For resistivity purposes, the dip is defined parallel to the horizontal resistivity (Harigawa, 1996). Interpretation of electrically anisotropic reservoirs from vertical resistivity, R_v , and horizontal resistivity, R_h has been covered extensively in the literature by Klein et al., 1997, Mollison et al., 1999, Shoen et al., 1999, Shray et al., 2001, Clavaud et al., 2005, Cao Minh, 2008 and many others.

Historically, wells were drilled vertically and logged with wireline tools. Conventional induction resistivity tools induce current loops that circulate perpendicular to the wellbore, and measure horizontal resistivity. Horizontal wells logged with wireline tools, or conventional EM resistivity tools create current loops that circulate in the vertical direction and measured vertical resistivity. With the addition of wireline triaxial resistivity tools in the late nineties, all three resistivity components were measured simultaneously.

Over the past twenty years, LWD EM tools have advanced from a relatively short transmitter-to-receiver spacing with a single tilted antenna, to the current technology architecture which has multiple collars each with several antennas tilted at 45 degrees that are non-collocated. These are often referred to as being pseudo-triaxial and pseudo-collocated as they have some sensitivity to multiple axis around the wellbore. (Seydoux et al, 2014) Their inversions provide maps of horizontal and vertical resistivity and dip at a specific inversion point. Understanding the accuracy and uncertainty of these inversions has mostly focused on the horizontal resistivity because the horizontal resistivity map is what is primarily used for geosteering.

Introduction

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Having an accurate near and far-field wellbore formation anisotropy will allow more advanced petrophysical analysis, such as for thin bed formations. (Manuaba et al, 2024). It will also remove the presence of geometrical effects (e.g., polarization, proximity, and shoulder bed), allowing for a more direct comparison to vertical or near-vertical offset wells. The data can also be used to discriminate between high anisotropy shale beds and low anisotropy water-filled clastic formations, for example. Incorporating this information into the real-time evaluation would allow for improved geosteering and reservoir characterization, leading to a more successful well.

This paper examines various types of anisotropy. In-layer (or near-field) anisotropy pertains to the intrinsic anisotropic properties within a layer, detectable during horizontal drilling. Boundary anisotropy emerges at the interface between two isotropic layers with distinct resistivities, where the resistivity contrast manifests as an anisotropic response in measurements. Far-field anisotropy occurs when the tool's detection depth makes it sensitive to adjacent layers, such as a shale layer beneath an isotropic formation. This concept is particularly advantageous for distinguishing between conductive layers, where shales exhibit anisotropic properties, and water-bearing zones are typically isotropic.

Understanding the origins of anisotropy within formations is essential. Shale formations are renowned for their intrinsic anisotropic properties due to their microscopic grain structure, resulting in significant electrical anisotropy. However, shales are not the only anisotropic layers; sandstones can also exhibit macroscopic anisotropy due to laminations or grain size variations. Transverse isotropy (TI) in shales arises from the microscopic grain structure, as noted by Hornby (1993). The horizontal layering of plate-like particles within shales leads to electrical anisotropy, where resistivity measured perpendicular to the bedding plane (R_v) may greatly exceed resistivity measured parallel (R_h), termed microscopic anisotropy.

Conversely, formations comprising laminations thinner than the spatial resolution of certain resistivity logging tools may exhibit effective TI anisotropy, despite individual isotropic laminates. In laminates composed of alternating conductive shales and resistive oil sands, R_h is predominantly influenced by shale, while R_v is determined by the resistive hydrocarbon zone. Consequently, R_h measurements alone lack sensitivity to hydrocarbons.

Anisotropy is of particular interest to operators aiming to determine water saturation, using equations such as Archie's and Indonesia, which relies on true resistivity (R_t). Low water saturation correlates with high reservoir resistivity, while high saturation results in low resistivity. In evaluating anisotropic formations such as turbidites, which consist of alternating layers of shales and sands, it is essential to consider the

resistivity characteristics R_v and R_h . Klein (1997) demonstrated that R_h is limited as an indicator of water saturation, whereas R_v exhibits significant sensitivity and is therefore a more reliable indicator of water saturation. Understanding these resistivity characteristics is important for accurate evaluation and resource management.

Methodology

From a physics perspective, the sensitivity of deep and ultra-deep EM measurements (commonly known as DAR and UDAR respectively) to formation resistivity and anisotropy depends on several factors including the relative formation resistivity, tool operating frequency, transmitter and receiver array spacing, relative tool position, and how the measurements are constructed. The terminology deep (DAR) refers to azimuthal resistivity measurements that measure (considering the previously mentioned factors) in the range of feet to tens of feet, whereas ultra-deep (UDAR) refers to a depth of detection in the order of tens to hundreds of feet (e.g., reservoir scale).

The most widely used deep EM measurements include harmonic resistivity, harmonic anisotropy, symmetrized, and anti-symmetrized channels. These measurements are constructed in such a way that they respond to petrophysical properties and formation geometry in a significantly different manner. Their unique responses not only offer a quick way to understand the formation surrounding the wellbore but also enable imaging the formation in a quantitative way via inversion workflows. This paper will focus on harmonic resistivity and harmonic anisotropy responses.

While harmonic resistivity channels are mainly intended for formation resistivity, they are a function of anisotropy and dip in addition to resistivity. For an array of a short spacing measurement operating at low frequencies, the phase channels are proportional to $1/R_h + 1/(R_v \xi)$, where $\xi = \sqrt{1 + (R_h/R_v - 1)\sin^2\theta}$ in homogeneous formations with a dipping anisotropy (Wang et al., 2025). Here, angle θ is anisotropy dip. In horizontal wells where the formation is also horizontally lying, the relationship reduces to $1/R_h + 1/\sqrt{R_h R_v}$.

Harmonic anisotropy channels exhibit a more complex relationship with formation properties. They are constructed using two coplanar couplings. In homogeneous formations with a dipping anisotropy, the phase channels are proportional to $1/R_h - 1/(R_v \xi) - 1/R_h (1 - \xi) (\cos^2 \theta + 1)/\sin^2 \theta$ for short spacing arrays at low frequencies (Wang et al., 2025). When the tool axis is perpendicular to the R_h plane, harmonic anisotropy channels collapse to zero, so they cannot always be used to determine formation resistivity as harmonic resistivity. When the tool axis is parallel to the R_h plane, one coplanar coupling parallel to the R_h plane contains information of both R_h and R_v ; the other contains only the information of R_h . Usually, harmonic anisotropy channels have strong sensitivity to both R_h and R_v .

In the above, discussion has been focused on phase shift channels. Comparatively, attenuation channels are not as sensitive to resistivity change as phase shift channels when the spacing is short and/or the frequency is low. On the other hand, when the spacing is long and/or the frequency is high, both phase shift and attenuation channels offer comparable sensitivity to resistivity change in the formation. Their responses to R_h , R_v and dip are highly non-linear as a result of multiple interactions of high-order eddy currents induced in the formation, and hence an explicit relationship is not warranted.

Forward modelling has been performed to fully quantify how the UDAR hardware architectures which are currently on the market are sensitive to anisotropy within the formation for both in-layer and within a shoulder bed.

To perform this task, the following assumptions were made as part of the setup. Without loss of generality, we use an ultra-deep azimuthal tool with one transmitter (Tx) and two receivers (Rx), as shown in Figure 1. The transmitter-to-receiver spacings are 50 ft and 100 ft, respectively. At each receiver, three antennas are tilted at 45 degrees from the tool axis to acquire the electromagnetic signals from different directions. The antennas are operated at frequencies of 2 kHz and 24 kHz.



Figure 1—Example of UDAR BHA with transmitter-to-receiver spacings of 50 ft and 100 ft. Antennas are tilted at 45 degrees from the tool axis, with up to 3 tilted antennas per sub.

Two scenarios are considered to model tool responses. We first look at a tool placed in an infinitely large homogeneous medium to study the in-bed tool sensitivity, as shown in Figure 2A. The formation has a horizontal resistivity, R_h , and vertical resistivity, R_v , with the tool axis being placed perpendicular to the direction of R_v . Note that the bed becomes isotropic when horizontal resistivity is equal to vertical resistivity. We simulate the tool responses with R_h of 2, 5, 10, 20, and 50 ohm-m and anisotropies of 1, 2, 4, 6, 10, 15, and 20.

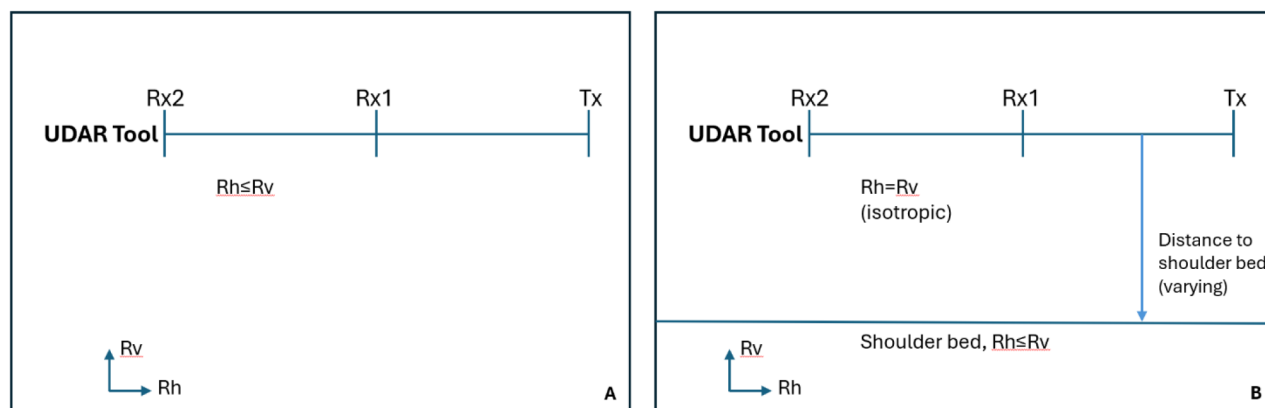


Figure 2—Synthetic models used to simulate tool responses in a horizontal well. The tool is shown in Figure 1. Figure A shows a 0D homogeneous medium whose resistivity and anisotropy vary within the modelling. Figure B shows a 1D, two-layer (with one boundary) formation, with the shoulder bed at a varying distance within the modelling. The in-layer medium is isotropic, and the shoulder-bed has a resistivity of 1 ohm-m with a varying anisotropy.

In the second scenario, we simulate the tool response in a two-layer formation, as shown in Figure 2B. The tool axis is parallel to the boundary to mimic a horizontal well. It is situated 25 ft away from the boundary. Afterwards, we also study the effect due to the distance between the tool and the shoulder bed, in which the proximity in the model changes from 50 ft to 0 ft above the boundary. The in-layer bed is an isotropic formation with varying resistivity, and the shoulder bed has a horizontal resistivity of 1 ohm-m but with varying anisotropy.

The first model can be categorized as zero-D (0D) and the second as one-dimensional (1D). It is worth noting that there may be scenarios in which a tool experiences a relative dip with respect to the R_h direction resulting in a higher dimensionality (2D or 3D) problem. However, this scenario is beyond the scope of this study as it would significantly increase the complexity of the tool's sensitivity.

Results

Figure 3 illustrates the difference in measurement response between the harmonic resistivity and harmonic anisotropy for tool firing frequencies of 2 kHz and 24 kHz in an anisotropic homogenous formation with varying resistivity and anisotropy ratio. Both scenarios illustrate the measurement response in a 0D formation and therefore can be described as illustrating their response for the in-layer anisotropy sensitivity. Generally, the dynamic ranges of these measurements are smaller at 2 kHz than at 24 kHz, indicating they are more sensitive to changes in resistivity and anisotropy ratio at higher frequencies. It is plausible that high-frequency measurements are preferable if the accuracy of formation properties is of higher priority than the depth of investigation. Moreover, at both frequencies, resistivity and anisotropy are better resolved at lower resistivity and anisotropy ratio than at the higher values. A change in either formation resistivity or anisotropy can be described by a change in conductivity. EM measurements are proportional to a small change in conductivity with respect to the background. When the change in anisotropy is due to the change in vertical resistivity, it is found that $\Delta d \sim \Delta \sigma \sim -1/((R_h \lambda^2) \Delta \lambda)$. Here, d is data, σ is conductivity, R_h and λ are horizontal resistivity and anisotropy ratio, respectively. This relationship shows that the higher the resistivity, the higher the anisotropy ratio, the smaller the data sensitivity is to anisotropy.

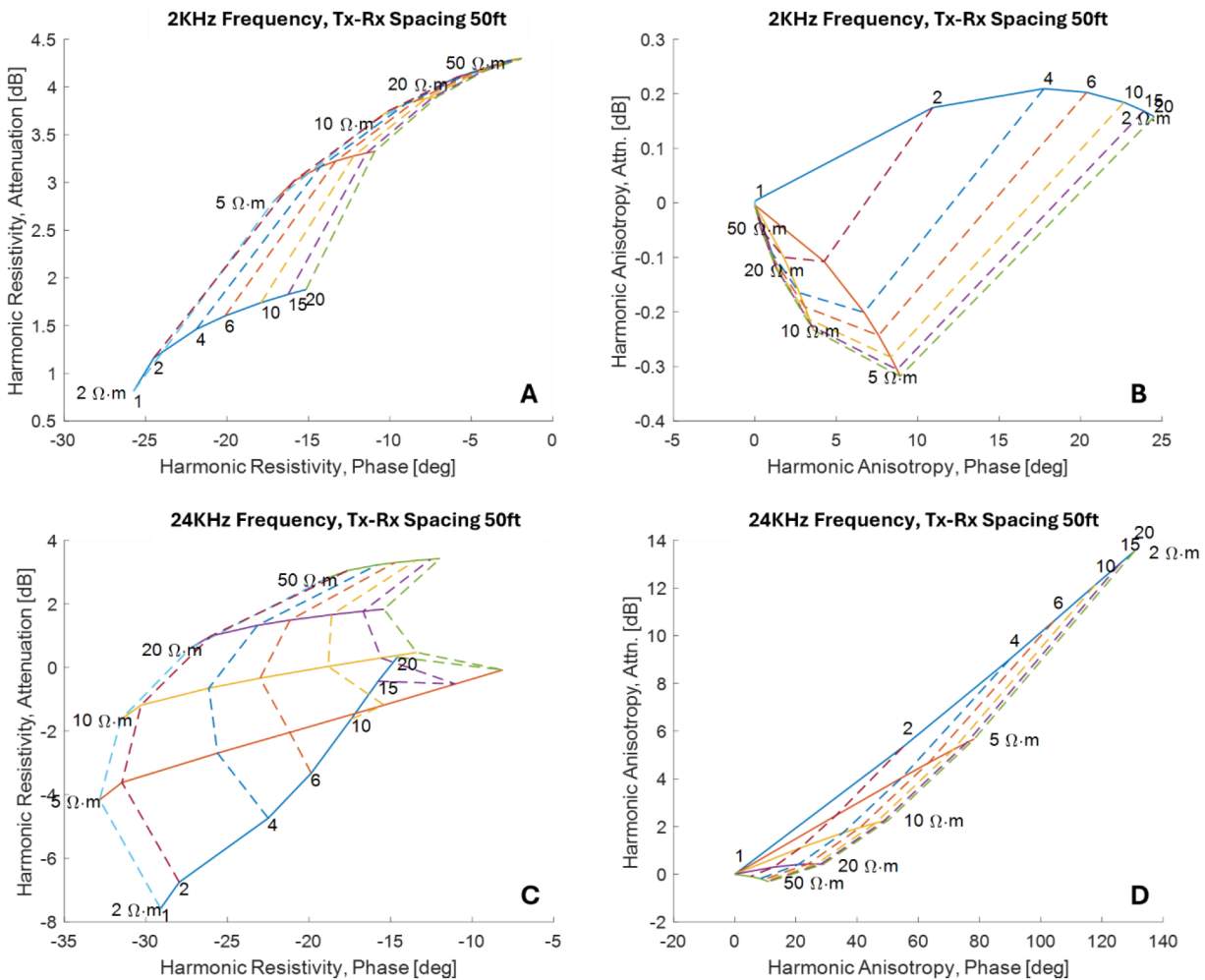


Figure 3—Highlighting the harmonic resistivity and harmonic anisotropy response in a 0D (no boundary) formation (see Figure 2A), referred to as in-layer sensitivity, with a transmitter-to-receiver spacing of 50 ft and with frequencies of 2 kHz and 24 kHz. The solid lines represent differing R_h values of the tool's medium, while the dotted lines illustrate the varying levels of R_v/R_h ratios. Note how the variance in the transmitter frequency changes the range of measurement response and sensitivity to anisotropy within the formation.

Figure 4 shows the measurement response of a 50 ft and a 100 ft long array, both using a frequency of 2 kHz. It is clear to see how different transmitter-to-receiver spacings yield different responses to resistivity and anisotropy. Apparently, an increase in spacing enhances sensitivity to both resistivity and anisotropy, similar to the effect of increasing frequency. Specifically, the dynamic ranges of phase shift and attenuation are greater at 100 ft spacing compared to 50 ft spacing across the entire ranges of resistivity and anisotropy under investigation. The most significant increase is observed on the attenuation of harmonic anisotropy, which shows substantially increased sensitivity to anisotropy where R_h is less than 5 ohm-m. In the meantime, sensitivity to cases where R_h is greater than 5 ohm-m is also improved, suggesting a better chance for increased accuracy in measuring resistivity and anisotropy.

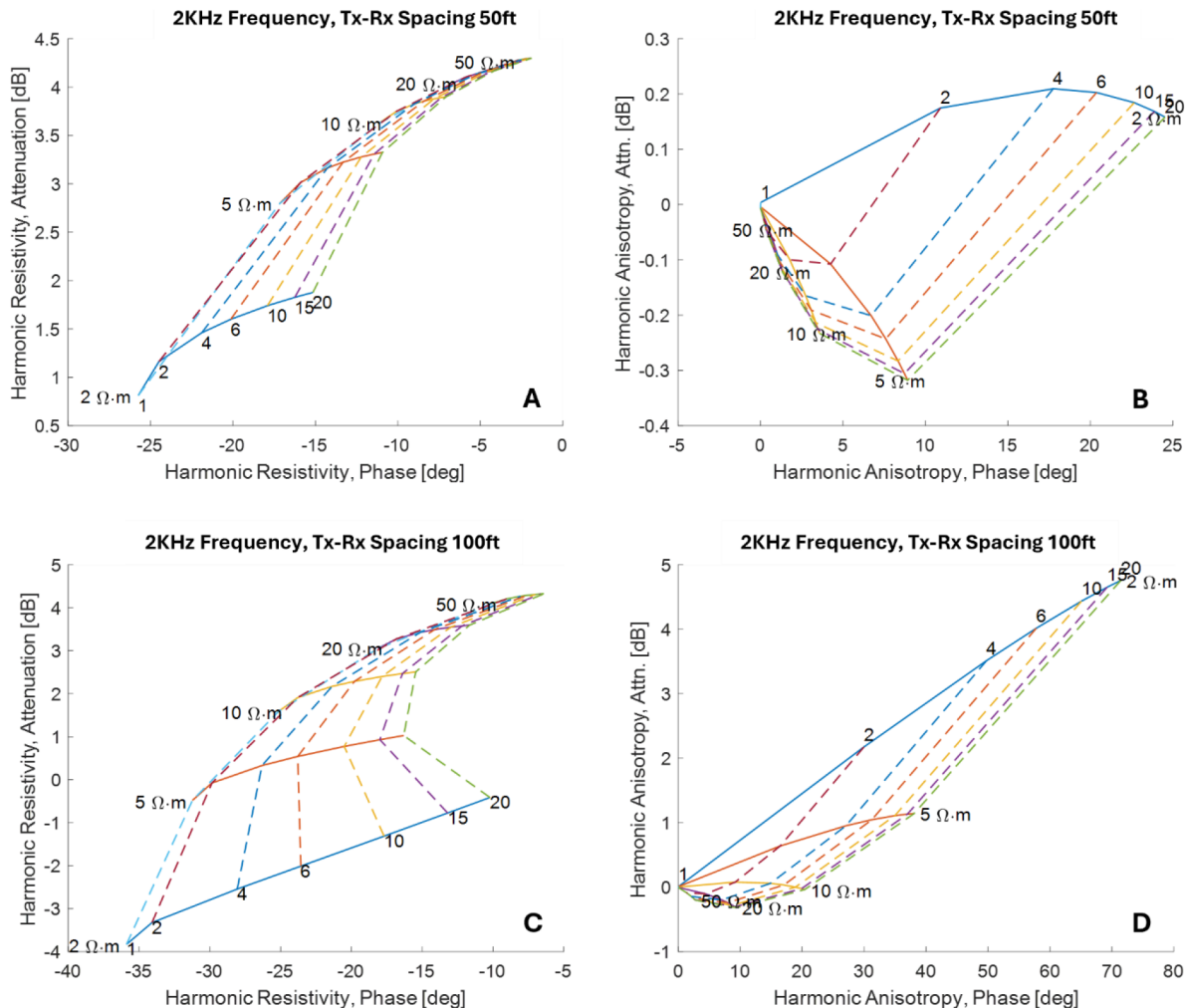


Figure 4—Highlighting the harmonic resistivity and harmonic anisotropy response in a 0D (no boundary) formation (see Figure 2A), referred to as in-layer sensitivity, with a transmitter-to-receiver spacing of 50 ft and 100 ft, all at 2 kHz frequency. The solid lines represent differing R_h values of the tool's medium, while the dotted lines illustrate the varying levels of R_v/R_h ratios. Note how the variance in the transmitter-to-receiver spacing changes the range of measurement response and sensitivity to anisotropy within the formation.

Results in Figures 3 and 4 illustrate that different frequencies and spacings give rise to varying sensitivities to resistivity and anisotropy. Frequency and spacing play similar roles in influence on sensitivity. Previous work shows that the most sensitive resistivity zone of a given array corresponds to the middle of the induction number range from 0.5 to 5 (Seydoux et al. 2014). In addition, the work suggests that optimal frequencies offering the best sensitivity exist for a given range of formation resistivity and a specific spacing. Results in Figures 3 and 4 suggest once optimal frequencies are determined for the target

range of resistivity, shifting the frequency spectrum to the higher end, where applicable, may help improve the sensitivity to anisotropy, which in turn may enhance the image quality of vertical resistivity.

Figure 5 highlights the UDAR tool's sensitivity to an anisotropic shoulder-bed environment, as described in Figure 2B, for the scenario where the tool is at a vertical distance of 25 ft TVD from the boundary. In this case, two spacings, 50 ft and 100 ft, are modeled with the operating frequency fixed at 2 kHz. Harmonic resistivity channels appear to be more sensitive to resistivity of the host bed than to anisotropy of the shoulder bed. The spread over anisotropy is small compared to the range of attenuation and phase shift over resistivity, and the change in transmitter-to-receiver spacing does not significantly alter this spread. Comparatively, harmonic anisotropy channels show stronger sensitivity to anisotropy, although their sensitivity to resistivity is not as strong as that of harmonic resistivity channels.

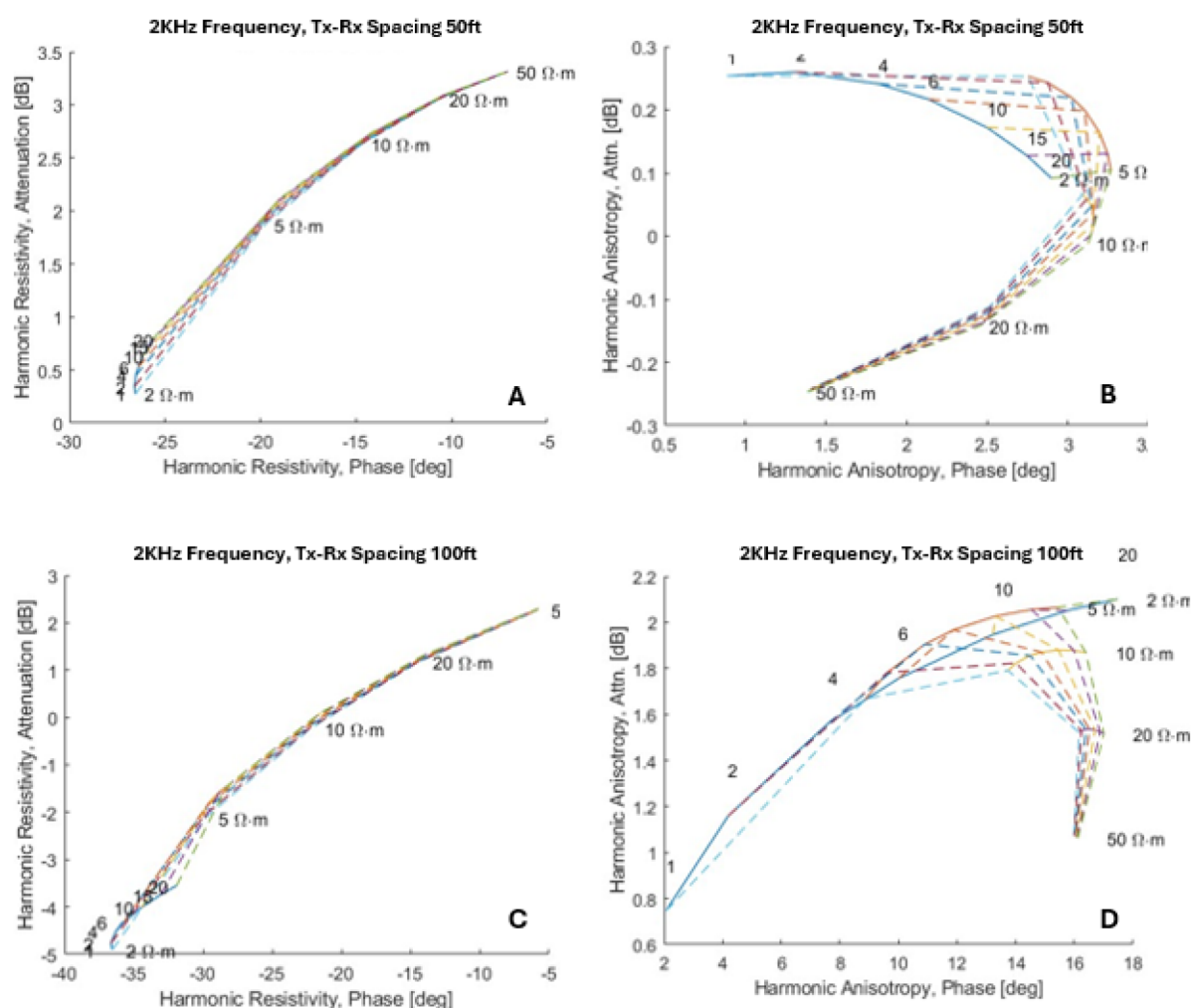


Figure 5—Highlighting the harmonic resistivity and harmonic anisotropy response in a two-layer (one boundary) formation (see Figure 2B), in this instance the shoulder bed is at a distance of 25ft TVD from the wellbore. The in-layer medium is isotropic, and the shoulder-bed has a R_h of 1 ohm.m. and a varying anisotropy ratio. All figures use a transmitter-to-receiver spacing of 50 ft but vary tool firing frequency of 2 kHz or 24 kHz. The solid lines represent differing R_h values of host bed, while the dotted lines illustrate its varying levels of R_v/R_h ratios of the shoulder bed. Note how the range of measurement response and sensitivity to anisotropy within the formation varies between transmitter-to-receiver spacing.

It is worth noting that the increase in spacing appears to significantly boost the sensitivity of harmonic anisotropy channels, in particular the phase shift channels, more than harmonic resistivity channels. In this two-layer case, harmonic anisotropy lives up to its name. While harmonic resistivity is crucial for determining in-layer resistivity due to its wide dynamic range and quasi-linear response, harmonic anisotropy is important for the accuracy of anisotropy of the shoulder bed. It is agreeable that the increased

sensitivity of harmonic anisotropy to the shoulder bed is the result of the greater depth of investigation of the longer spacing. On the other hand, results in the homogeneous case suggest that increased spacing itself can directly enhance the sensitivity. It seems plausible to conclude that both the longer spacing and the greater depth of investigation contribute to enhanced sensitivity.

Figure 6 clearly illustrates how proximity to the shoulder bed and resistivity of the host bed leads to a varying response of the UDAR measurements, based on the model seen in Figure 2B. The frequency used here is 2 kHz in all four sub-figures and two spacings, i.e. 50 and 100 ft, respectively are modeled in the study. Figures 6A and 6B represent the 50 ft spacing, while Figures 6C and 6D represent the 100 ft spacing. On the left (6A and 6C), harmonic resistivity attenuation is displayed, and on the right (6B and 6D), harmonic anisotropy attenuation for each spacing. The cyan-colored curves correspond to the isotropic shoulder bed, and the orange-colored curves to the anisotropic shoulder bed. The solid and dotted curves are obtained in the 2 ohm·m conductive host bed, while the dashed and dash-dotted curves are obtained in the 20 ohm·m resistive host bed.

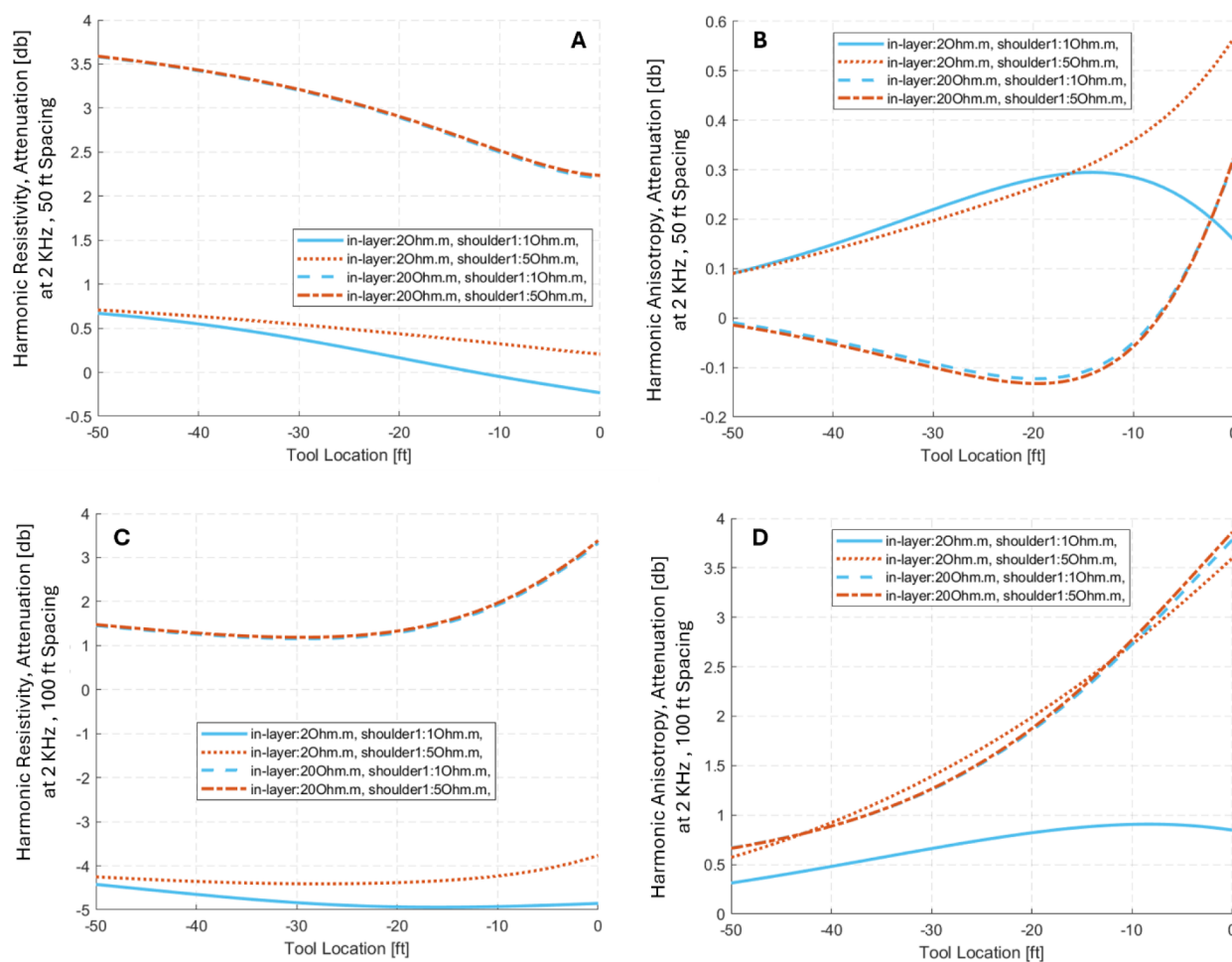


Figure 6—Highlighting the harmonic resistivity and harmonic anisotropy response in a two-layer (one boundary) formation (see Figure 2B), the shoulder bed has a varying distance from the wellbore. The in-layer medium has either a 2 ohm·m or 20 ohm·m isotropic characteristic. The shoulder-bed has a R_h of 1 ohm·m and an anisotropy ratio of 1:1 or 1:5. All figures use a tool firing frequency of 2 kHz, with a transmitter-to-receiver spacing of either 50 or 100 ft, respectively.

The most noticeable phenomenon across all four subplots is the clear separation of the two curves for the more conductive host bed. Moreover, this separation becomes increasingly large as the tool approaches the shoulder bed. In stark contrast, the two curves for the resistive host bed overlay each other at most distances until the tool is within 10 ft, where the only two curves that barely separate are the harmonic anisotropy

curves of the 100 ft spacing array. This phenomenon appears counter-intuitive because EM waves penetrate deeper in the resistive background, which suggests better sensitivity to the shoulder bed. However, it is consistent with simulation results in [Figures 3 and 4](#), where sensitivity to anisotropy is apparently better when the host bed is conductive ($R_h < 5 \text{ ohm}\cdot\text{m}$).

All curves show increased sensitivity to the position of the bed boundary as the tool gradually moves towards the shoulder bed. Comparatively, the sensitivity of harmonic anisotropy appears stronger than that of harmonic resistivity, as evidenced by the more rapid change in the former with respect to distance. This difference suggests a deeper depth of investigation for harmonic anisotropy compared to harmonic resistivity. Although the resistive host bed does not increase sensitivity to the anisotropy of the shoulder bed, the sensitivity to the position of the bed boundary is better when the tool is in the resistive bed than in the conductive bed. The behaviors of harmonic resistivity and harmonic anisotropy suggest that the accuracy of vertical resistivity R_v returned by the inversion may be better when the tool is in the conductive bed.

Discussion

All the results illustrated here logically follow the fundamental principles of electromagnetic propagation resistivity physics for UDAR measurements. They are that, given the current UDAR antenna architecture of 45 deg tilted antennas, the sensitivity is a function of transmitter-to-receiver spacing, tool firing frequency and the resistivity context. However, the geosteering answer of "it depends" is very pertinent in understanding exactly what a UDAR hardware's sensitivity to a formations anisotropy truly is.

Near-field (In-layer):

Firstly, it is important to think about what type of anisotropy we are trying to resolve for. [Figures 3 and 4](#) illustrated how both harmonic resistivity and harmonic anisotropy have good sensitivity to the in-layer anisotropy, but this is driven largely by how resistive that local medium is. When using relatively shorter spacings (e.g. 50 ft) a relatively higher frequency is most useful to characterize the anisotropy (see [Figure 3C and 3D](#) exhibiting a larger signal compared with [Figures 3A and 3B](#)). In this scenario, good sensitivity is achieved when R_h is up to approximately 20 $\text{ohm}\cdot\text{m}$. However, once the R_h exceeds this value, it is rapidly reducing a UDAR's sensitivity to the in-layer anisotropy.

When looking at the longer spacing antenna pairings (100 ft), the relatively lower firing frequency will deliver more sensitive UDAR measurements. [Figures 4A to 4D](#) highlight how the UDAR's overall measurement sensitivity to the in-layer's anisotropy significantly reduces in environments over 10 $\text{ohm}\cdot\text{m}$, and this sensitivity begins to taper meaningfully after the R_h is above 5 $\text{ohm}\cdot\text{m}$ when using a shorter spacing of 50 ft ([figure 4A and 4B](#)).

These results illustrate how the longer spacing UDAR measurements are less useful in capturing the sensitivity of an in-layer anisotropy given these spacings are usually paired with the lower frequency range. However, they will still add value to understanding the environment, especially in the more distal regions from the wellbore and particularly to the presence of a boundary.

Far-field (Shoulder bed):

[Figure 5](#) illustrates how very little anisotropic signal is present in the UDAR's measurements once additional boundaries are introduced to the environment. As outlined above, the R_h of the in-layer medium controls the sensitivity to the environment and this is still true for understanding the sensitivity to the shoulder bed's characteristics. Again, UDAR sensitivity to the shoulder bed anisotropy is significantly reduced once the R_h of the host medium is greater than 5 $\text{ohm}\cdot\text{m}$. To further understand the UDAR's sensitivity to the shoulder bed's anisotropy characteristics, proximity to its boundary is a major factor.

[Figure 6](#) reiterates the results discussed above, as [Figures 6A–6D](#) all show that a UDAR tool is insensitive to whether the shoulder bed is isotropic or anisotropic when in an in-layer R_h of 20 $\text{ohm}\cdot\text{m}$. This is evidenced by the dashed cyan- and dot-dashed, orange-colored lines tracking together, irrespective of proximity to the shoulder bed. It does show, however, that the harmonic resistivity and harmonic anisotropy are sensitive to

the boundary's presence, with a strong deflection in the curves present from up to approximately 20 ft TVD from the UDAR tool. This response still contributes to the geosteering purpose, however, does not provide information on a far-field environment's anisotropic characteristics.

In contrast, it is clear to see the difference in the response of shoulder bed anisotropy when the tool is located within an in-layer medium of just 2 ohm·m (solid cyan- and dotted, orange-colored curves exhibit a noticeable divergence). Furthermore, the UDAR's measurement response is significantly more sensitive to the shoulder bed's anisotropy characteristics at distance when a transmitter-to-receiver spacing of 100 ft is acquired (Figure 6D). However, when a smaller UDAR spacing is employed, (50 ft, Figure 6B), sensitivity to the shoulder bed's anisotropic characteristics is only achieved once within 15 ft proximity.

Inverted outputs:

Ultimately, understanding what sensitivities the current UDAR hardware has within its measurements, will help the end users to process and then interpret their inverted results. Having now seen how the measurement sensitivity is affected by factors including in-layer resistivity and shoulder bed anisotropy, it is clear to then recognize these effects on the inverted outputs.

By utilizing a fully automatic, Deterministic Parametric Inversion (as outlined in Habashy and Abubakar, 2004, Thiel et al, 2012) which inverts for an individual layer, R_h and R_v , it is possible to utilize the UDAR's sensitivity to anisotropic environments to produce an inverted output for use in a real-time geosteering operation, or when deployed in a more passive reservoir mapping context.

Figures 7 to 10 illustrate exactly how the sensitivity carried in the UDAR measurements translate into inverted results. In these four figures, each has a scenario where a UDAR toolset with spacings of 50 and 100 ft is within a borehole with an angle of ~90deg and is at a true vertical distance (TVD) of 45 ft, 25 ft and 5 ft above the shoulder bed. The models in Figures 7 and 8 both have an in-layer isotropic resistivity of 2 ohm·m, with Figure 7 shoulder bed having a R_h of 1 ohm·m and a R_v of 1, while Figure 8 shoulder bed has a R_h of 1 ohm·m and a R_v of 5. In contrast, Figures 9 and 10 both have an in-layer isotropic resistivity of 20 ohm·m, with Figure 9 shoulder bed having a R_h of 1 ohm·m and a R_v of 1, while Figure 10 shoulder bed has a R_h of 1 ohm·m and a R_v of 5 (see Table 1 below).

Table 1—Formation parameters for Figures 7 through 10.

	In-layer Isotropic Resistivity (ohm·m)	Shoulder bed R_h (ohm·m)	Shoulder bed R_v (ohm·m)
Figure 7	2	1	1
Figure 8	2	1	5
Figure 9	20	1	1
Figure 10	20	1	5

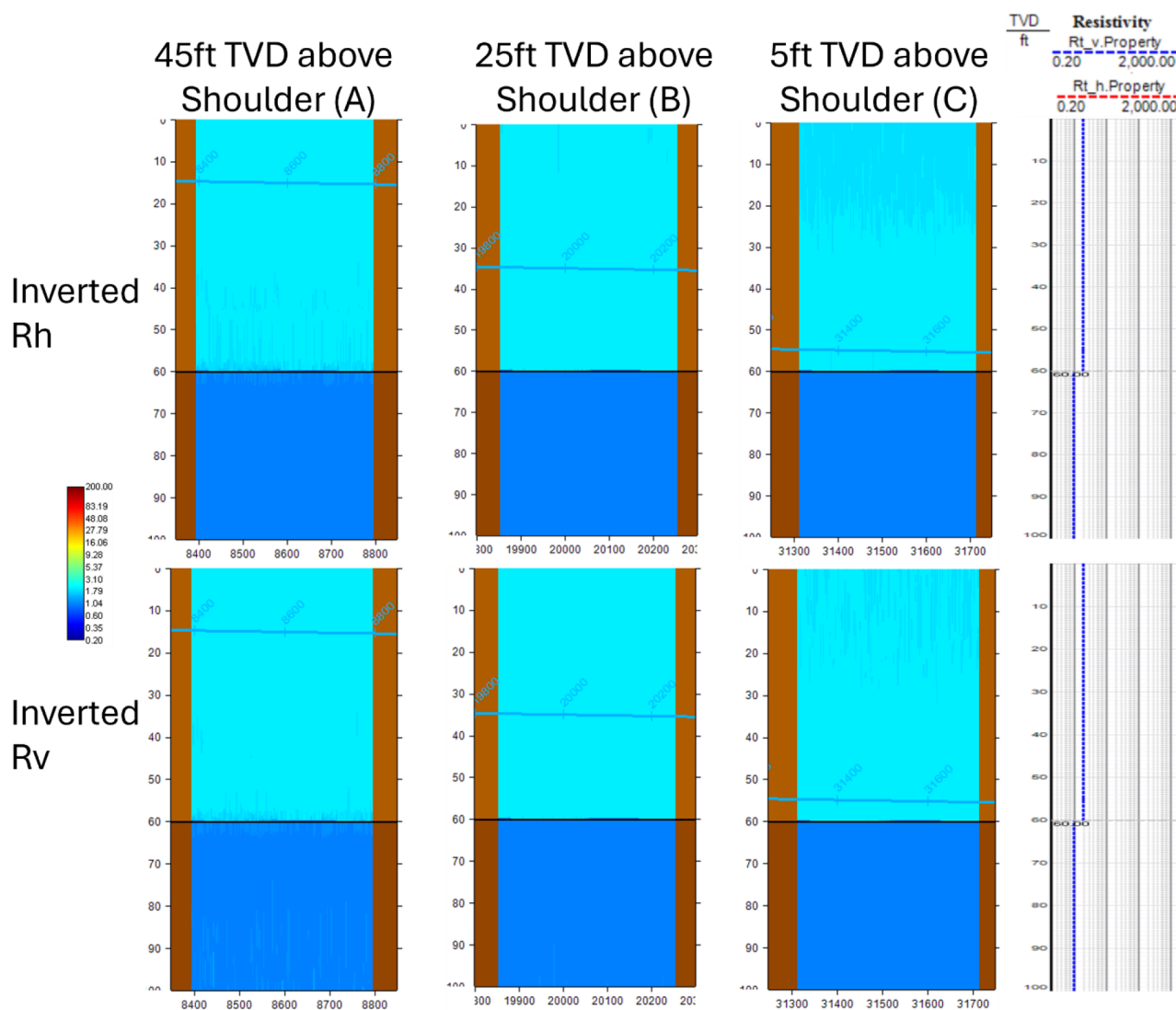


Figure 7—1D DPI inversion results using forward modeled UDR measurements. For each sub-image, the uppermost image shows inverted R_h and the lowermost image shows inverted R_v , at TVD 45 ft(A), 25 ft(B), 5 ft(C) above the shoulder bed. Each image shows 400 ft MD and the well inclination (in blue) is $\sim 90^\circ$. True formation model is seen in vertical log to the right; in-layer resistivity: 2 ohm.m, shoulder bed: 1 ohm.m with anisotropy: 1.

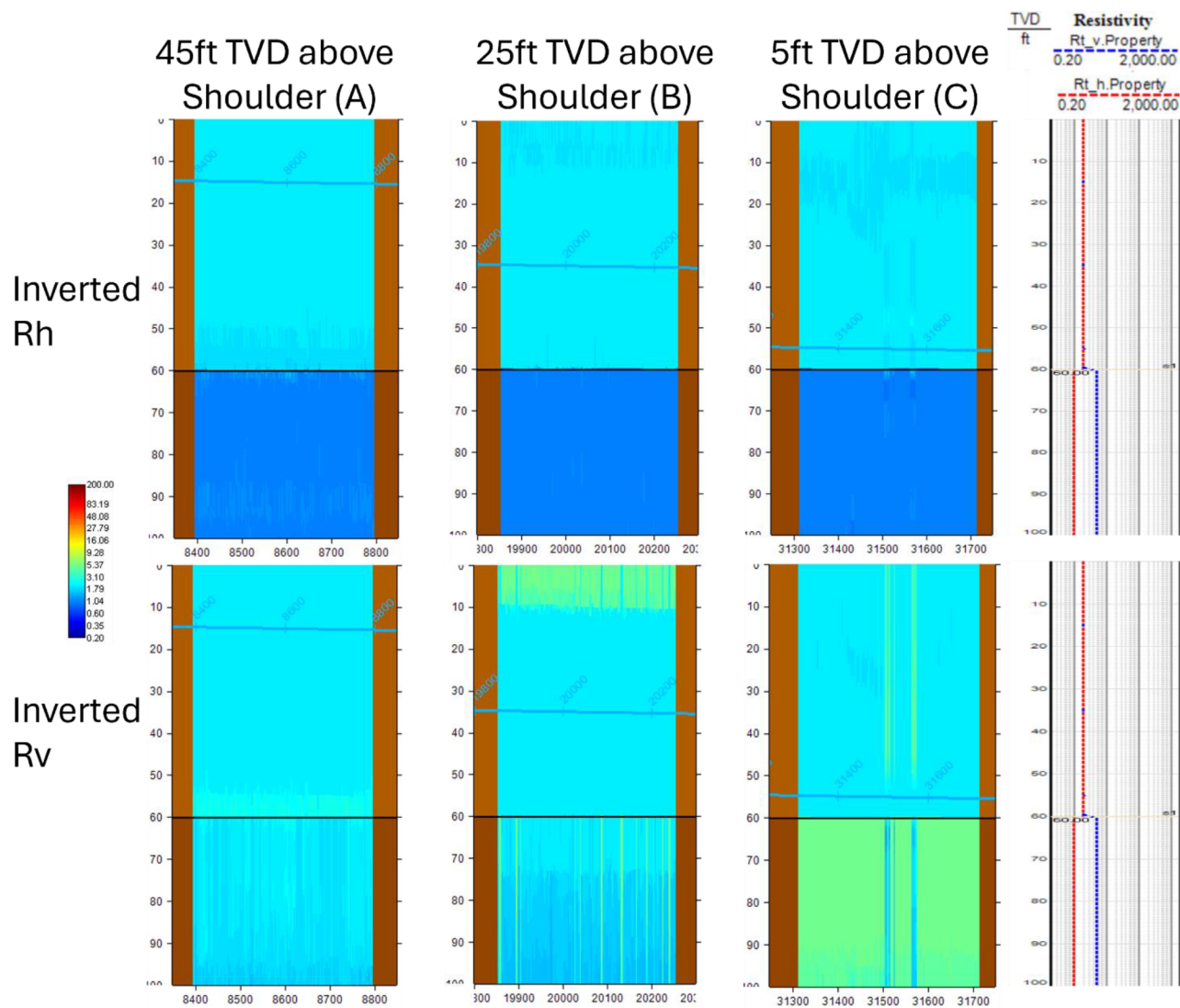


Figure 8—1D DPI inversion results using forward modeled UDR measurements. For each sub-image, the uppermost image shows inverted R_h and the lowermost image shows inverted R_v , at TVD 45 ft(A), 25 ft(B), 5 ft(C) above the shoulder bed. Each image shows 400 ft MD and the well inclination (in blue) is ~90deg. True formation model is seen in vertical log to the right; in-layer resistivity: 2 ohm.m, shoulder bed: 1 ohm.m with anisotropy: 5.

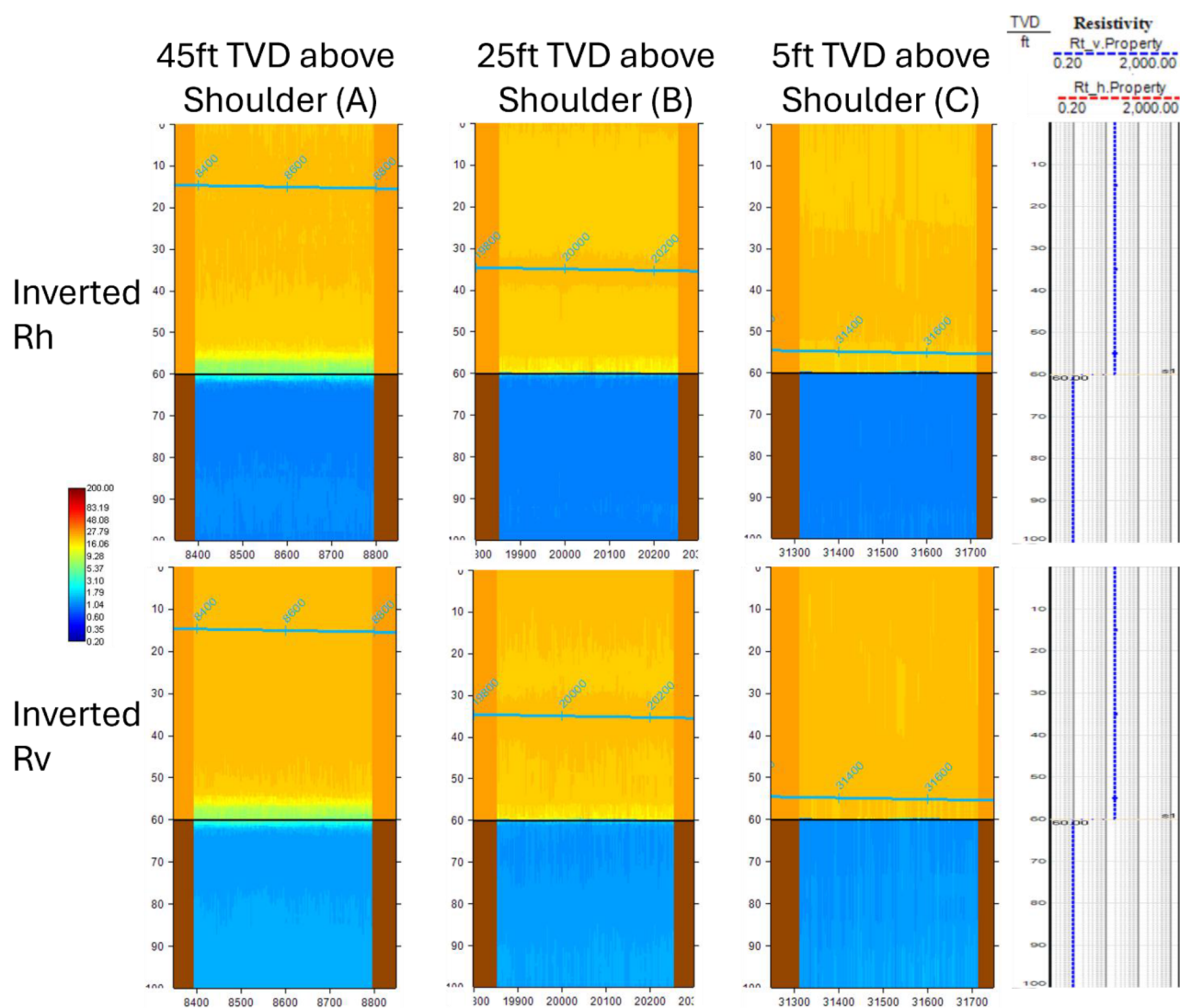


Figure 9—1D DPI inversion results using forward modeled UDR measurements. For each sub-image, the uppermost image shows inverted R_h and the lowermost image shows inverted R_v , at TVD 45 ft(A), 25 ft(B), 5 ft(C) above the shoulder bed. Each image shows 400 ft MD and the well inclination (in blue) is $\sim 90^\circ$. True formation model is seen in vertical log to the right; in-layer resistivity: 20 ohm.m, shoulder bed: 1 ohm.m with anisotropy: 1.

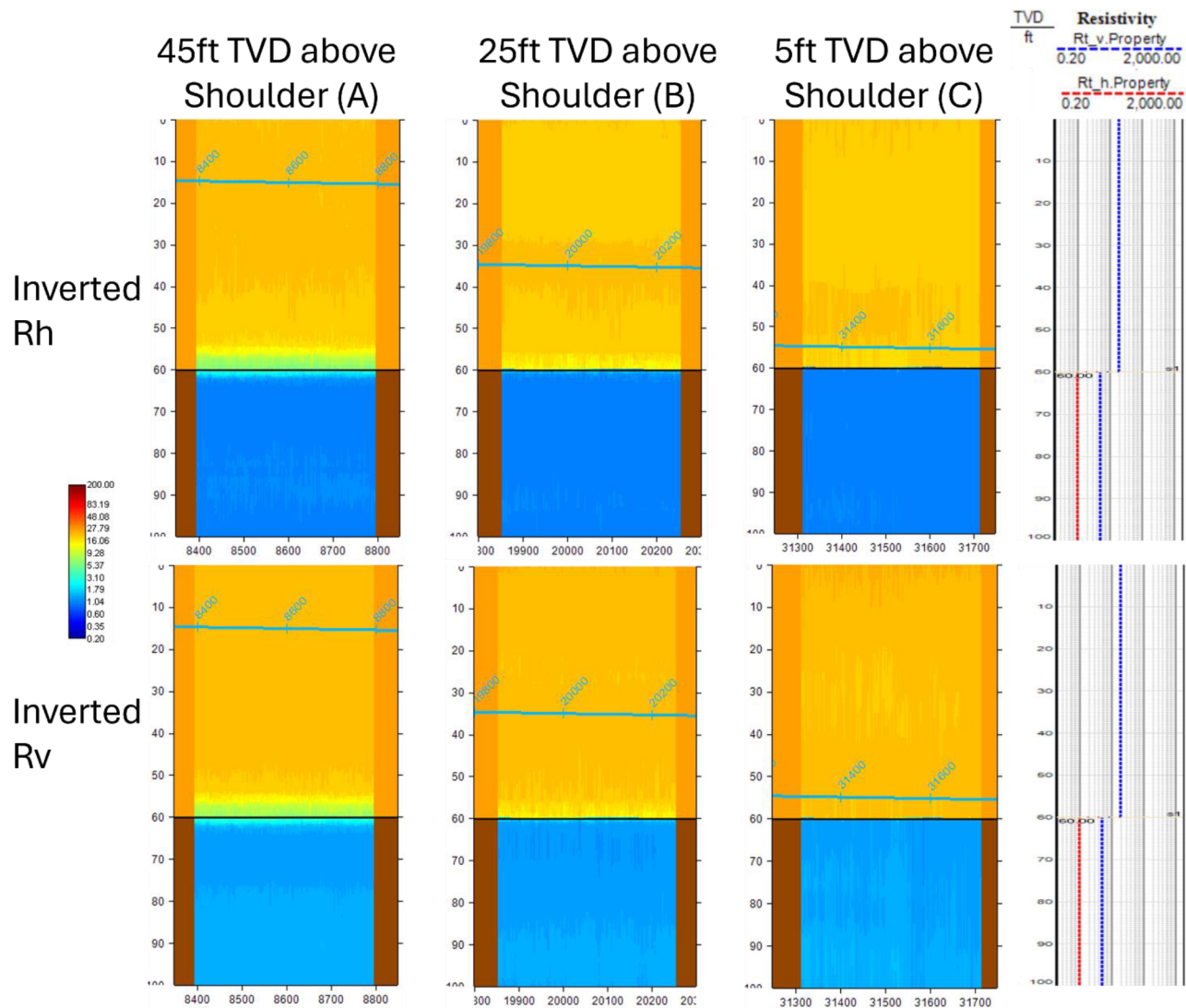


Figure 10—1D DPI inversion results using forward modeled UDR measurements. For each sub-image, the uppermost image shows inverted R_h and the lowermost image shows inverted R_v , at TVD 45 ft(A), 25 ft(B), 5 ft(C) above the shoulder bed. Each image shows 400 ft MD and the well inclination (in blue) is $\sim 90^\circ$. True formation model is seen in vertical log to the right; in-layer resistivity: 2 ohm.m, shoulder bed: 1 ohm.m with anisotropy: 5.

Figures 7 and 9 both illustrate the correct inverted R_h and a R_v , all of which are isotropic in nature, in both the in-layer medium and the shoulder bed, no matter the distance from the shoulder bed. They both also invert the boundary position correctly, highlighting the results in Figure 6 that the UDR tool carries sensitivity to a boundary's presence, even when both layers are isotropic. Although it is worth noting in the 45 ft TVD scenarios, that the confidence in the inverted boundary position is reduced slightly due to the distal nature of the shoulder bed boundary.

It can be seen in Figure 8A however, that even at 45 ft TVD above the anisotropic shoulder bed, the UDAR measurements carry some sensitivity to the anisotropy present, as the inversion is able to invert for the anisotropic shoulder bed, albeit at a R_v of 2 ohm.m. and not 5 ohm.m. as is the truth in the model. This sensitivity is likely contributing from the harmonic anisotropy channels, which show sensitivity to shoulder bed anisotropy while in a 2 ohm.m in-layer medium, with the 100 ft spacing and 2 kHz frequency (Figure 6D). As the UDAR approaches the shoulder bed the sensitivity to the anisotropic shoulder bed increases slightly (Figure 8B), indicated by the increasing inverted R_v values of 2.2-2.5 ohm.m, and then by the time the wellbore is at 5 ft TVD the inverted R_v is correct, reconciling the true R_v of 5 ohm.m.

Lastly, Figure 10 emphasizes the findings from Figures 5 and 6, that when a UDAR tool with this tilted antenna architecture is in a resistive in-layer medium (in the case of Figure 10, is 20 ohm·m), there is little-to-no sensitivity to the anisotropy of a shoulder bed carried in the UDAR measurements. Figures 10A, 10B and 10C all manage to correctly invert for the shoulder bed's boundary position (Figure 10A with an increase uncertainty as in Figure 9A), however they all fail to invert for a R_v of more than 1.3 ohm·m even at a distance of 5 ft TVD (Figure 10C). This departure from an inverted R_v of 1 ohm·m is likely from the uncertainty in the inversion results (Bower et al, 2025), rather than the UDAR measurements carrying a small amount of sensitivity to the anisotropy of the shoulder bed.

Consequences on Geosteering:

What does this mean for the practical application of UDAR technology? Given that the primary purpose of the technology is to geosteer in real-time and/or map the reservoir in a more passive manner, then it is important to characterize if it is lithological or fluid characteristics which are driving the UDAR's response. A key method to differentiate between these two is to identify the presence of anisotropy from lithological factors (e.g. shale) or the presence of isotropy from fluid-fill factors (e.g. water filled sand). This is often conveyed by utilizing a 1D, 2D or even 3D inversion, since there is a rich measurement data set available, and interpretation of the individual measurements alone is not practically possible.

The results presented in this paper, highlight for the first time that, to understand the anisotropic characteristics of a shoulder bed is going to be extremely challenging, even when using sophisticated inversion algorithms. There is a narrow set of scenarios where the UDAR architecture does have some sensitivity to a far-field anisotropy characteristics, mainly limited to an environment with an in-layer R_h of less than 5 ohm·m.

The results also illustrate the importance of robust and thorough forward modelling. This is crucial during the pre-job stage to optimize the set-up of real-time array spacing and frequencies. Additionally, it remains essential to the post-job stage as it is key to validating an interpretation or hypothesis and is critical for closing the loop.

Way forward

This paper illustrates how it is possible to select a transmitter-to-receiver spacing and firing frequency to ensure the maximum sensitivity possible to the resistivity environment, however, ultimately the current antenna design is limited in how much sensitivity is achievable. Fundamentally, the current non-collocated 45 deg tilted antenna design yields a moderate signal-to-noise characteristic and the inability to fully decouple sensitivity to individual measurement axis. This limits the architecture's ability to deliver even a qualitative measure of anisotropy, except for in a narrow range of scenarios.

The need for new technologies to be able to quantitatively measure near- and far-field anisotropies is clear. With an increasing focus on water injectors to stimulate mature fields, the ability to measure anisotropy in very conductive environments is important to maximize reservoir contact. In hydrocarbon reservoirs, with the exception of low resistivity pay, the ability to quantitatively differentiate whether an inverted conductive response (at a distance from the wellbore) is due to an isotropic water-filled reservoir versus an anisotropic shale is beyond the current generation of UDAR architecture and measurements.

Any new tool architecture will need to increase the UDAR's sensitivity by utilizing technology that is currently accepted as the industry standard and conveyed on wireline, namely a fully triaxial and collocated antenna architecture.

Conclusions

The results presented here show that there is no simple answer to how much sensitivity the current UDAR tool architectures have to near-field or far-field anisotropy. The modelling utilized an EM propagation tool, which is representative of the common tilted antenna architecture currently available. The modelling utilized scenarios with varying resistivities, anisotropies levels and shoulder bed relative positions with respect to the

tool, to quantify exactly in what scenarios the harmonic resistivity and harmonic anisotropy measurements had sensitivity to the change in model parameters.

It is illustrated that the current UDAR architecture has sensitivity to the near-field anisotropy (also known as in-layer), however the main control of its sensitivity is the R_h of the bed, for a given spacing and firing frequency. The modelling also found that the tool's sensitivity reduces rapidly once the resistivity value exceeds 20 ohm-m, with both the harmonic resistivity and harmonic anisotropy having a good range of sensitivity below that.

The research also illustrated how the sensitivity to a shoulder bed's anisotropy, also termed far-field sensitivity, is significantly reduced when compared to near-field sensitivity, for the equivalent measurements. A main control on what sensitivity is present is again the R_h of the host bed, and if this value is greater than 5 ohm-m, then the UDAR hardware in its current form has little-to-no sensitivity of the far-field anisotropy. In fact, by an in-layer R_h of 20 ohm-m, the UDAR is totally insensitive to the anisotropic characteristics of the far-field, irrespective of the UDAR's tool relative position. Although it can be clearly seen the measurements still react to the presence of a boundary, regardless of the anisotropic characteristics.

It is noteworthy however, that tool's proximity to the shoulder bed is a secondary control on the UDAR's sensitivity to that bed. If the tool is in a low resistivity (sub-5 ohm-m) layer, then the harmonic resistivity and the harmonic anisotropy could have sensitivity to the anisotropy present more than 45ft away.

All these findings have been reinforced with the measurements being input to a high-resolution deterministic inversion, which is capable of inverting for individual layer anisotropy. Its results illustrated that when the UDAR tool is in an in-layer resistivity of 20 ohm-m, it cannot differentiate the presence of far-field anisotropy. Conversely, the inversion conveys the measurement's sensitivity to the anisotropic shoulder bed, when the tool is in an in-layer R_h of 2 ohm-m.

In all scenarios, it is important to use appropriate spacings and frequency inputs, along with ensuring that good quality (low noise) input data is selected. Using as high frequency as possible for the given spacing is key to enable maximum sensitivity to any anisotropic character.

The findings will have a significant impact on how this information is used for both geosteering and reservoir mapping scenarios. To be able to increase the sensitivity to the anisotropy of the formation at both near-field (in-layer) and far-field (shoulder bed), it is proposed that a new type of LWD measurement would be required. This would require an improved measurement quality when compared to the current architecture, along with a fully-triaxial and collocated antenna, akin to that currently accepted as the industry standard on a wireline conveyance.

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